

Large improvement in trap level and space charge distribution of polypropylene by enhancing the crystalline – amorphous interface effect in blends

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Abstract

The accumulation of space charge in high voltage direct current cable will bring the insulation to failure; the addition of nanoparticles can markedly improve the space charge distribution characteristics inside the cable insulation, but particle agglomeration and cavitation lead to difficulty in controlling the properties of nanocomposites. In this paper, polypropylene (PP)/propylene-ethylene copolymer (PEC) and PP/ethylene-octene copolymer (EOC) blends were prepared by mechanical blending in order to improve both mechanical properties and space charge distribution. Dynamic mechanical thermal analysis shows that both blends have excellent mechanical properties for recyclable power cable. Pulsed electro-acoustic and thermally stimulated depolarization current tests illustrate that PP/EOC blends significantly decrease space charge accumulation and remarkably increase the trap density in the bulk compared with PP and PP/PEC blends. The increase of the trap density in PP/EOC blends can be explained as the result of the shallow traps introduced by the crystalline – amorphous interface existing in the boundaries of spherulites. The shallow traps can act as hopping sites to improve the transportation of space charges.

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Keywords: crystalline – amorphous interfaces; trap level distribution; space charge distribution; blend

INTRODUCTION

Extruded cable with crosslinked polyethylene (XLPE) insulation has been successfully applied to high voltage direct current (HVDC) power cables, due to its favourable combination of superior mechanical, thermal and electrical characteristics. However, XLPE is hard to recycle because of the thermosetting property when the cable reaches the end of its life. This results in a waste of resources and great damage to the environment. Thus minimizing the environmental destruction of XLPE by developing recyclable power cable insulation materials is significantly important.

Some work has been done on polyethylene and polypropylene as environmental electrical insulation for next generation extruded cables. Previous research on low density polyethylene (LDPE) and its blends,¹ high density polyethylene (HDPE) and its blends² and LDPE/HDPE blends³ indicates that ethylene-based systems do not have sufficient mechanical properties at high temperature and cannot be operated at temperatures exceeding 90 °C. Isotactic polypropylene (PP), which has superior thermal properties to polyethylene, is a promising material for large-capacity HVDC power cables. However, its inherent rigidity, especially at lower temperature, is a barrier to utilization as the insulation of power cables. Recently, to enhance the toughness of PP, a new family of metallocene-catalysed elastomers such as propylene – ethylene copolymers (PECs) and ethylene – octene copolymers (EOCs) has attracted great attention due to their narrow molecular mass distribution, making them easier to handle and allowing faster mixing. It has been found that PP/PEC and PP/EOC blends as cable insulation having higher operating temperature

than LDPE and HDPE can provide very good heat deformation performance.^{4,5}

Apart from the mechanical property, which is of vital importance, there is a key requirement for outstanding space charge distribution for HVDC cable. The accumulation of space charge in HVDC cables is a tough problem that brings about insulation failure.^{6,7} Recently, many researchers have reported that the addition of nanoparticles can markedly improve the space charge distribution characteristics.^{8–10} Nevertheless, incompatibility and insufficient interfacial adhesion between the particle and the polymer are usually great challenges and they may cause particle agglomeration and cavitation in composites, which in some cases have detrimental effects on the performance of the insulation.¹¹

Blending is a simple and practical method to modify the space charge distribution in polymers. Up to now, space charge behaviours of blend polymers of LDPE have been widely studied.^{3,12–14} However, the mechanism of space charge suppression in blend polymers has not yet been unambiguously explained since blending not only changes the morphologies of the individual semicrystalline polymers but also generates the interfaces between different phases of the two polymers. It is

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widely accepted that space charge behaviour in an insulating polymer is closely related to carrier traps introduced by interfaces between crystalline and amorphous material.¹ The carrier traps existing in the crystalline–amorphous interfaces of polymer blends are a complex function of interfacial types and interfacial area. The former can be divided into three interfaces, including lamella and amorphous regions, matrix and elastomers, spherulite boundaries. The latter depends on the degree of crystallinity, elastomer dispersion and spherulite size. At present, the understanding of the origin of carrier traps in crystalline–amorphous interfaces of semicrystalline polymer blends is still very limited.

In this paper, PEC and EOC were blended with PP by mechanical blending. The space charge profiles of PP and its blends were investigated by pulsed electro-acoustic (PEA) tests. The mechanism to suppress the tested space charge is discussed on the basis of the trap level distribution calculated by thermally stimulated depolarization current (TSDC) measurements. In addition, the origin of carrier traps in crystalline and amorphous interfaces was investigated by SEM, TEM, wide-angle X-ray diffraction (WAXD), small angle X-ray scattering (SAXS), DSC and polarized optical microscopy (POM).

EXPERIMENTAL

Materials and preparation of specimens

Materials

Isotactic polypropylene T30s (Maoming Petrochemical Company, Guangdong, China) with a melt flow index of 3 g/10 min, density of 0.9 g cm⁻³, was used as the base material. PEC Vistamaxx 6102 (Exxon Mobil Chemical Company, TX, USA) and EOC Engage 8842 (Dow Chemical Company, MI, USA) were used to toughen PP. Their properties are listed in Table 1.

Preparation of specimens

PP was dried at 100 °C for 24 h in a vacuum oven before blending. PP/PEC and PP/EOC blends were prepared by melt-blending in a Hapro mixer. A cavity temperature of 453 K and a rotor speed of 60 rpm were used. The concentration of the PEC and EOC in both blends was 40 wt%. After blending, all samples were manufactured by hot pressing. The films were pre-melted by applying a curing press at 200 °C for 10 min, pressed under a pressure of *ca* 20 MPa for 10 min, and then pressed at 25 °C under a pressure of *ca* 5 MPa for a further 10 min.

Materials characterization

Scanning electron microscopy (SEM)

To observe the phase structure of PEC and EOC in the PP matrix, SEM images of the etched cross-section were examined at 10 kV in an ultra-high resolution Zeiss Sigma instrument. The freeze-fractured and etched cross-sections were coated with gold prior to examination under the electron beam.

Transmission electron microscope (TEM)

A JEOL 2011 transmission electron microscope, operated at 200 kV, was used to characterize the interface between the PP and elastomers. Samples were sectioned with a diamond knife on a Leica ultracut cryo-ultramicrotome to approximately 70 nm thickness at a sample temperature of -80 °C, and then the sections were collected on carbon-coated copper grids and stained with OsO₄ for 30 min to enhance compositional contrast between phases.

Table 1. Properties of PEC and EOC

Properties	PEC	EOC
Ethylene content (wt%)	16	86.4 ¹⁵
Density	0.862	0.857
Elongation at break (%)	>2000	>600

Dynamic mechanical thermal analysis (DMTA) measurements

DMTA was conducted with a TA Q800 instrument. The investigated samples of 20 mm × 10 mm area were cooled to -90 °C and heated to 90 °C at a heating rate of 3 °C min⁻¹ in a flow of nitrogen under a strain-controlled sinusoidal tensile load at a frequency of 1 Hz and a strain of 0.01%.

Differential scanning calorimetry (DSC) analysis

The crystallinity of the samples was examined using DSC (TA Q2000 instrument). For each test, a sample of *ca* 5 mg was first heated to 200 °C at 10 °C min⁻¹ and then annealed for 5 min to eliminate any residual nuclei and to ensure an identical thermal history. The sample was subsequently subjected to a heating/cooling rate of 10 °C min⁻¹ between 20 and 200 °C in a nitrogen atmosphere.

Polarized optical microscopy (POM) observation

The size of the spherulites was studied using POM. POM observation was performed using a Nikon Eclipse LV100NPOL microscope equipped with a crossed polarizer and a hot stage (Linkam LTS420). To observe the crystal morphology, the specimen was first melted at 200 °C and kept at this temperature for 10 min, and then quenched to the crystallization temperature ($T_c = 135$ °C) and maintained at T_c for the time necessary for isothermal crystallization.

Wide-angle X-ray diffraction (WAXD) tests

A D8 Advance diffractometer (Rigaku, Japan) equipped with a Cu K_α tube was employed to investigate the crystalline form of the samples over a range of diffraction angles $2\theta = 10^\circ - 30^\circ$.

Small angle X-ray scattering (SAXS) tests

The SAXS experiments were performed using a Rigaku SAXS instrument with a Cu K_α radiation source over a range of scattering angles $2\theta = 0.06^\circ - 1.5^\circ$.

Trap level distribution

The trap level profiles in PP and PP blends were calculated by the modified thermally stimulated depolarization current (TSDC) method.¹⁶ Gold electrodes with a diameter of 20 mm were sputtered on both sides of the samples for TSDC measurements. The sample was first polarized at 20 °C with a positive dc electric field of 3 kV mm⁻¹ for 20 min and then cooled at a rate of 10 °C min⁻¹. After the sample was cooled to -100 °C, the poling voltage was moved and the sample was shorted for 10 min. The depolarization current was measured from -100 to 100 °C at a heating rate of 3 °C min⁻¹.

Space charge measurements

The space charge measurements were carried out with the PEA system equipped with a Lecroy WaveRunner 610Zi (1 GHz, 10 GS

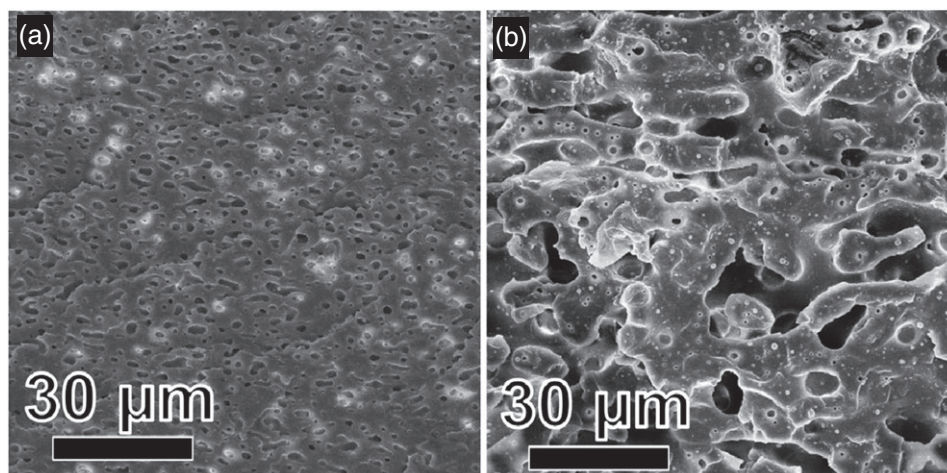


Figure 1. SEM images exhibit the elastomer dispersibility and cross-section microstructure in (a) PP/PEC blends and (b) PP/EOC blends.

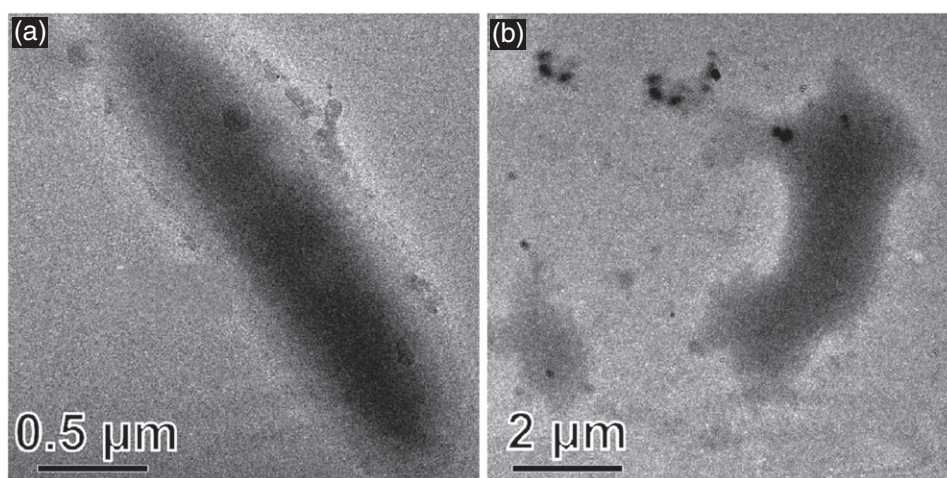


Figure 2. TEM image of sections stained with OsO_4 : (a) PP/PEC blend; (b) PP/EOC blend.

s^{-1} digital oscilloscope). A dc electrical field of 60 kV mm^{-1} and an electric pulse with an amplitude of 800 V and duration of 7 ns were applied to the plate samples for 1800 s. A piezoelectric transducer of polyvinylidene fluoride film with a thickness of $9 \mu\text{m}$ was used to change the detected acoustic signal into an electrical signal. Calibration for overshoot due to the nonlinear phase distortion was completed by signal processing. Meanwhile, calibration for attenuation was finished to obtain accurate information about the space charge profiles.¹⁷

RESULTS AND DISCUSSION

Dispersion and morphology

It is well known that elastomer dispersion and particle size in the matrix are the key factors to control the properties of blends. Figure 1 presents the dispersions of PEC and EOC in PP. It can be observed that both blends have a sea-island morphology, where the sea is the PP phase and the islands are the elastomer phase. No sharply defined interfaces between different domains can be seen in Fig. 1, which indicates good miscibility of the PP/PEC and PP/EOC blends. In addition, the holes in the PP/EOC blends are much bigger than those in the PP/PEC blends. On the one hand, this indicates that PP/PEC blends have better compatibility than PP/EOC blends; on the other, it suggests that more interfaces

exist between the matrix and elastomer in PP/PEC blends than in PP/EOC blends.

In order to characterize the interface between the PP matrix and elastomer phases clearly, TEM images (Fig. 2) were obtained in addition to SEM images. The staining agent OsO_4 cannot react with PP, and only elastomer phases are stained dark. The images clearly demonstrate the presence of an interphase layer at the PP–elastomer interface, which originates from the high compatibility between PP and elastomers. The interphase layer is composed of amorphous elastomers and semicrystalline PP.¹⁸

Dynamic mechanical thermal analysis

The HVDC cable is generally installed at ambient temperature and operates at high temperature. Therefore it is necessary to focus on the mechanical properties at room temperature and high temperature. Both the storage modulus and the glass transition temperature (T_g) estimated from $\tan \delta$ versus temperature are effective methods to assess the mechanical properties of polymers.

Figure 3(a) describes the values of the storage modulus as a function of temperature. At room temperature, blending PP with the new novel elastomers PEC or EOC results in a remarkable decrease

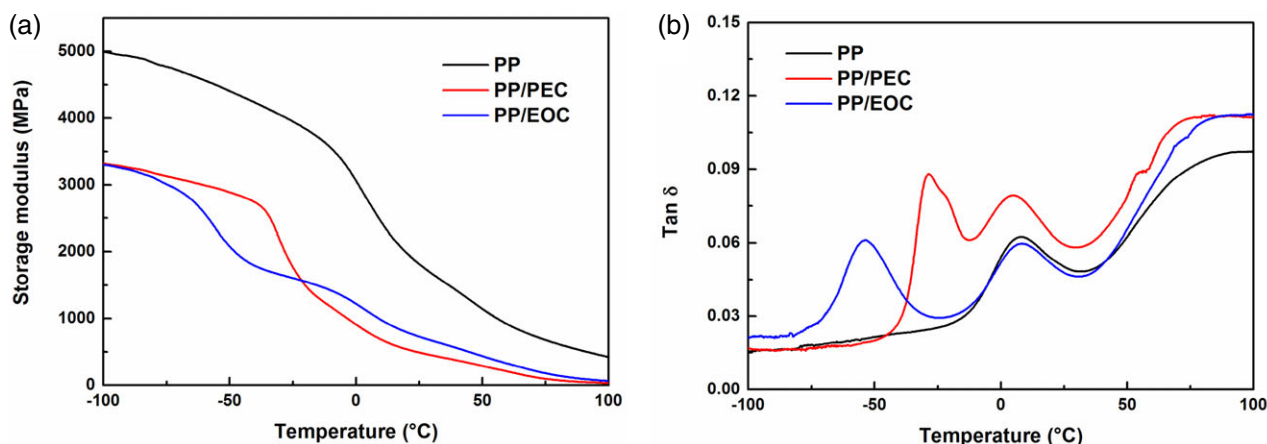


Figure 3. Storage modulus (a) and $\tan \delta$ curves (b) of PP and PP blends.

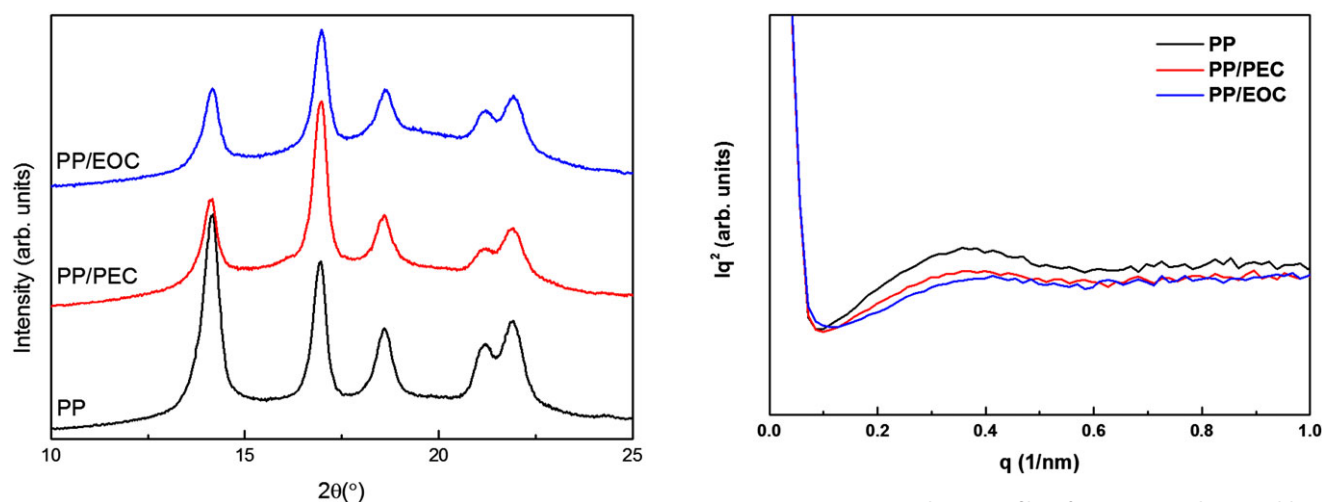


Figure 4. XRD patterns of PP, PP/PEC and PP/EOC blends.

in storage modulus, providing an effective way to improve the impact resistance of PP. Meanwhile, both PP/PEC and PP/EOC blends exhibit a significantly improved modulus at 120 °C compared with LDPE, which shows very poor mechanical properties above 60 °C.²

The brittleness of PP at low temperature is due to the relatively high T_g . From the $\tan \delta$ curves in Fig. 3(b), one clearly resolved peak is evident for PP itself at about 16 °C attributed to the glass transition of the PP amorphous phase, consistent with a previous study.¹⁹ For both PP/PEC and PP/EOC blends, two features were observed in the $\tan \delta$ curves. A feature at high temperature derives from the glass transition of the PP phase, and a second feature at low temperature associates with the glass transition of the elastomer phase.

On the basis of the dynamic mechanical analysis, a concentration of 40 wt% PEC or EOC added to PP is thought to be a suitable incorporation for power cable insulation.

Crystallization characteristics

In order to study the crystalline structure of PP and its blends systematically, WAXD, SAXS, DSC and POM analyses were performed.

Figure 5. Lorentz-corrected SAXS profiles of PP, PP/PEC and PP/EOC blends.

Crystal structure analysis

In the case of isotactic PP there are three well-known crystal structures: monoclinic (α), trigonal (β) and triclinic (γ) depending on the melting history, crystallization temperature, pressure and cooling rate as well as the presence of a blending material. Figure 4 shows X-ray diffractograms of the PP and PP blends. All samples have five main crystalline diffraction peaks at 2θ values of 14.1°, 16.8°, 18.4°, 21.0° and 36.1°, corresponding to the 110, 040, 130, 131 and 111 lattice planes respectively of the monoclinic crystalline form of PP. No trigonal or triclinic form refraction was found in the X-ray diffractograms. Therefore the crystallinity type of the two blends is the same as pure PP.

Figure 5 shows the Lorentz-corrected SAXS profile $I(q)q^2$ versus q where $q = 4\pi \sin \theta / \lambda$ (θ is the scattering angle, λ the wavelength) of PP and PP/PEC and PP/EOC blends. One scattering peak characteristic originating from the average thickness of the lamellar stacks (long period)¹⁸ is observed at $q^* = 0.37 \text{ nm}^{-1}$ for PP, PP/PEC and PP/EOC blends. It indicates that addition of the elastomers has no obvious effect on the lamellar thickness of the PP matrix.

Non-isothermal melt crystallization behaviours

Figure 6 presents the melting and crystallization behaviours of the PP and PP blends and Table 2 summarizes the crystallization temperature T_c , melting enthalpy ΔH_m and crystallinity X_c of the

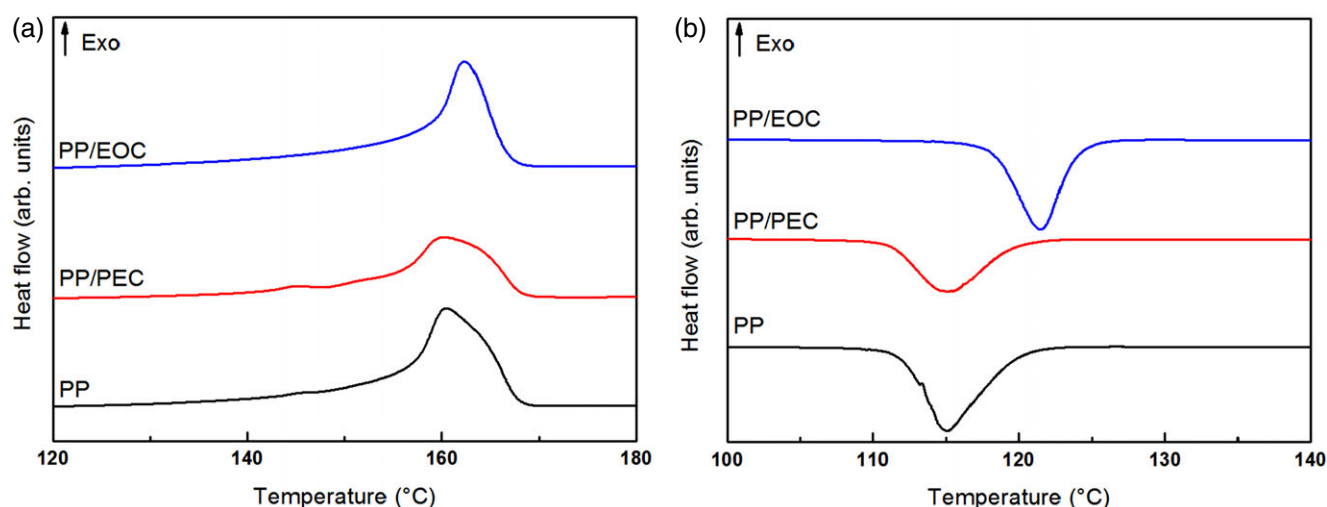


Figure 6. Melting (a) and non-isothermal crystallization (b) curves of PP and the PP blends.

Table 2. Crystallization temperature and crystallinity (wt%) of PP and PP blends

Materials	T_c (°C)	ΔH_m (J g ⁻¹)	X_c (wt%)
PP	115.1	93.7	44.8
PP/PEC	115.5	69.0	33.0
PP/EOC	121.5	84.5	40.4

blends. Both the virgin polymer and its blends display one feature, a sharp endothermic melting process at about 160 °C. It is clear that incorporation of elastomer cannot basically change the T_m of neat PP. Therefore both PP/PEC and PP/EOC blends appear to be candidates for HVDC cable, at least from the point of view of their melting behaviour.

The crystallinity of PP and its blends was also studied. The crystallinity degree was calculated from melting enthalpies according to

$$X_c = \Delta H_m / \Delta H_m^\infty \quad (1)$$

where ΔH_m is the fusion enthalpy of the blends and ΔH_m^∞ is the melting enthalpy of the completely crystalline PP. A value of 209 J g⁻¹ for 100% crystalline PP homopolymer was taken from

literature values.^{20,21} The crystallinity of PP in PP/EOC blends is 40.4%, which is 7.4% higher than that of PP/PEC blends and 4.4% lower than that of the pure PP. The crystallinity reflects the total interfaces between crystalline and amorphous material. This means that PP has more total interfaces between crystalline and amorphous material than its blends.

From the non-isothermal crystallization curves of the pure PP and its blends, it is found that both crystallization peak and onset temperature of crystallization of PP/EOC shift towards higher temperature compared with pure PP and PP/PEC, implying that EOC is a very effective nucleating agent. These results may be explained by the PP molecular chains aggregating to the long EOC molecular chains and crystallizing.^{22,23} The crystallization peak of PP/PEC blends has not been obviously improved, proving that the PEC phase has little effect on the nucleation of the blend.

Crystalline morphology

To deepen the understanding of the crystallinity characteristics of pure PP and PP/PEC and PP/EOC blends, polarized micrographs were investigated, as shown in Fig. 7. Virgin PP shows the typical perfect α -spherulites and the average diameter of the PP spherulites is about 50 to 80 μm . For PP/PEC blends, an almost

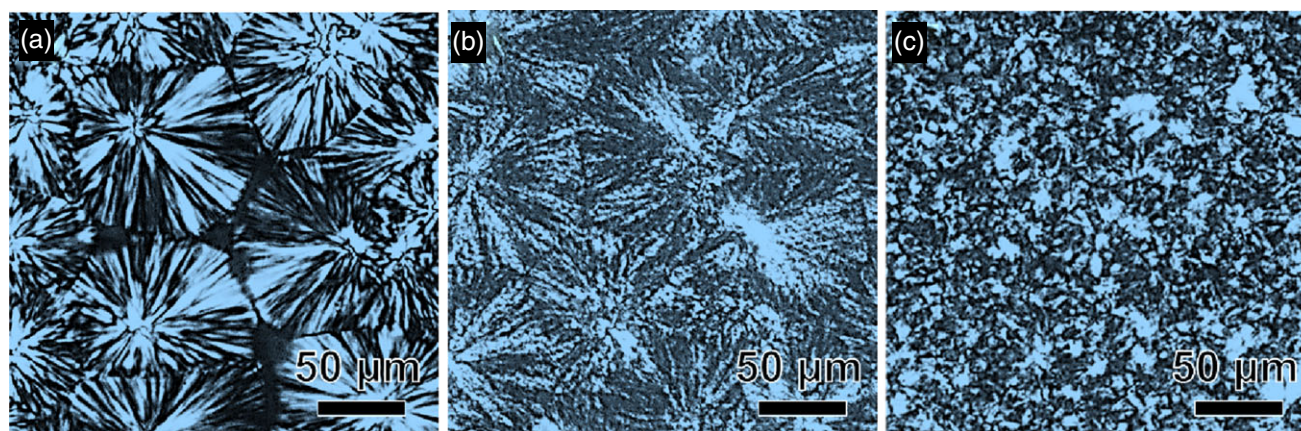


Figure 7. POM images show the isothermal crystallization morphologies of (a) PP, (b) PP/PEC blend and (c) PP/EOC blend.

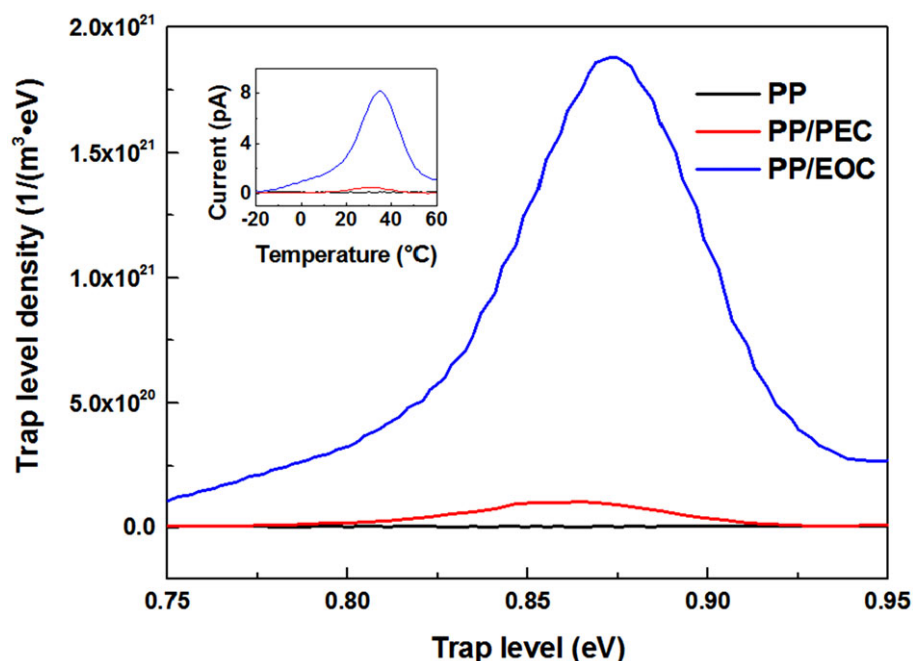


Figure 8. Trap level distribution in PP and PP blends; inset, the thermally stimulated current spectrum.

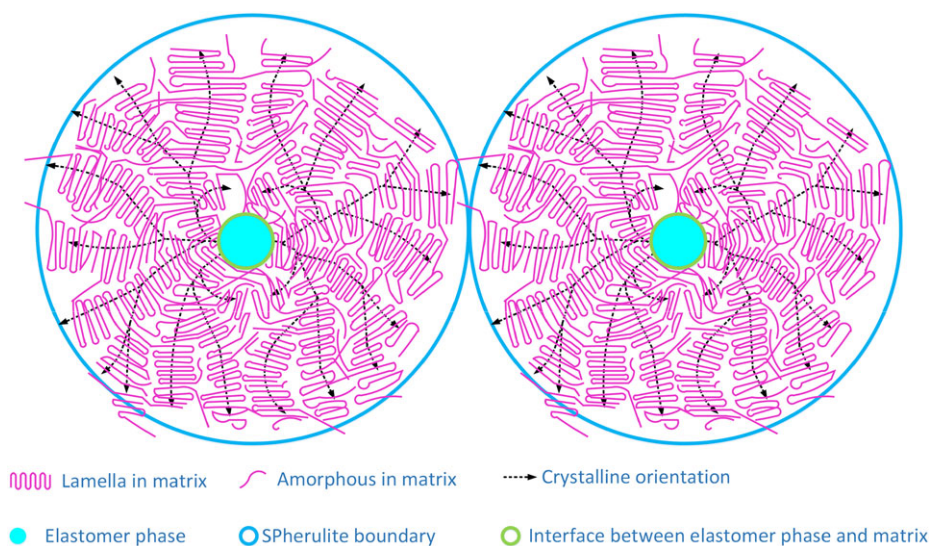


Figure 9. Interfaces in PP/PEC and PP/EOC blends.

identical diameter of α -spherulites is observed, further suggesting the weak nucleation effect of PEC particles in accordance with non-isothermal melt crystallization behaviours in PP/PEC blends. For PP/EOC blends, no spherulite structure can be seen in the POM micrograph (Fig. 7(c)) upon addition of EOC, indicating the excellent nucleation effect of EOC on PP crystallization by dramatically increasing the nucleation density and significantly reducing the spherulite diameters. The results are consistent with previous studies.^{22,24} The change in crystalline morphology should have a remarkable effect on the trap level density of the samples. This effect will be discussed later.

Trap level distribution

Based on TSDC theory, the trap level distribution can be calculated from the TSDC spectrum. The trap level distributions of PP and

PP blends are shown in Fig. 8 by using the modified TSDC theory. The thermally stimulated current spectrum is an inset in Fig. 8. The results show that the trap levels are both in the range 0.8–1.0 eV with a peak density at 0.87 eV, which can be regarded as shallow trap levels in PP.²⁵ The peak of the shallow trap level density in PP/EOC blends is about $1.88 \times 10^{21} \text{ m}^{-3} \text{ eV}^{-1}$, which is 17.7 times the peak in PP/PEC blends ($1.06 \times 10^{20} \text{ m}^{-3} \text{ eV}^{-1}$).

Interfaces between crystalline and amorphous material are thought to be the key factor to generate carrier traps of semicrystalline polymers.^{1,3,13,14} There are three main interfaces between crystalline and amorphous material, including lamella and amorphous regions (I), matrix and elastomer (II) and spherulite boundaries (III), as illustrated in Fig. 9. The interface types of the investigated samples are listed in Table 3.

Table 3. The interface types of PP and PP blends

Material	Interfaces
PP	Lamella and amorphous Spherulite boundaries
PP/PEC	Lamella and amorphous PP and PEC
PP/EOC	Spherulite boundaries Lamella and amorphous PP and EOC Spherulite boundaries

At present the understanding of the origins of carrier traps in semicrystalline polymer blends is still very limited. SEM and POM images can semi-quantitatively reflect the area of interfaces II and III. The crystallinity calculated from DSC tests reflects the total interfaces between crystalline and amorphous. PP, which only has interfaces I and III, has mostly interface I area and least interface III area. Combining the negligible carrier traps in PP (as shown in Fig. 8) compared with its blends indicates that interfaces I do not result in many carrier traps. PP/PEC blends have nearly identical interface III area, but more interface II area and less interface I area than PP, demonstrating that interfaces II can generate a few traps. The SEM and POM images illustrate that PP/EOC blends have more interfaces III and fewer interfaces II than PP/PEC blends,

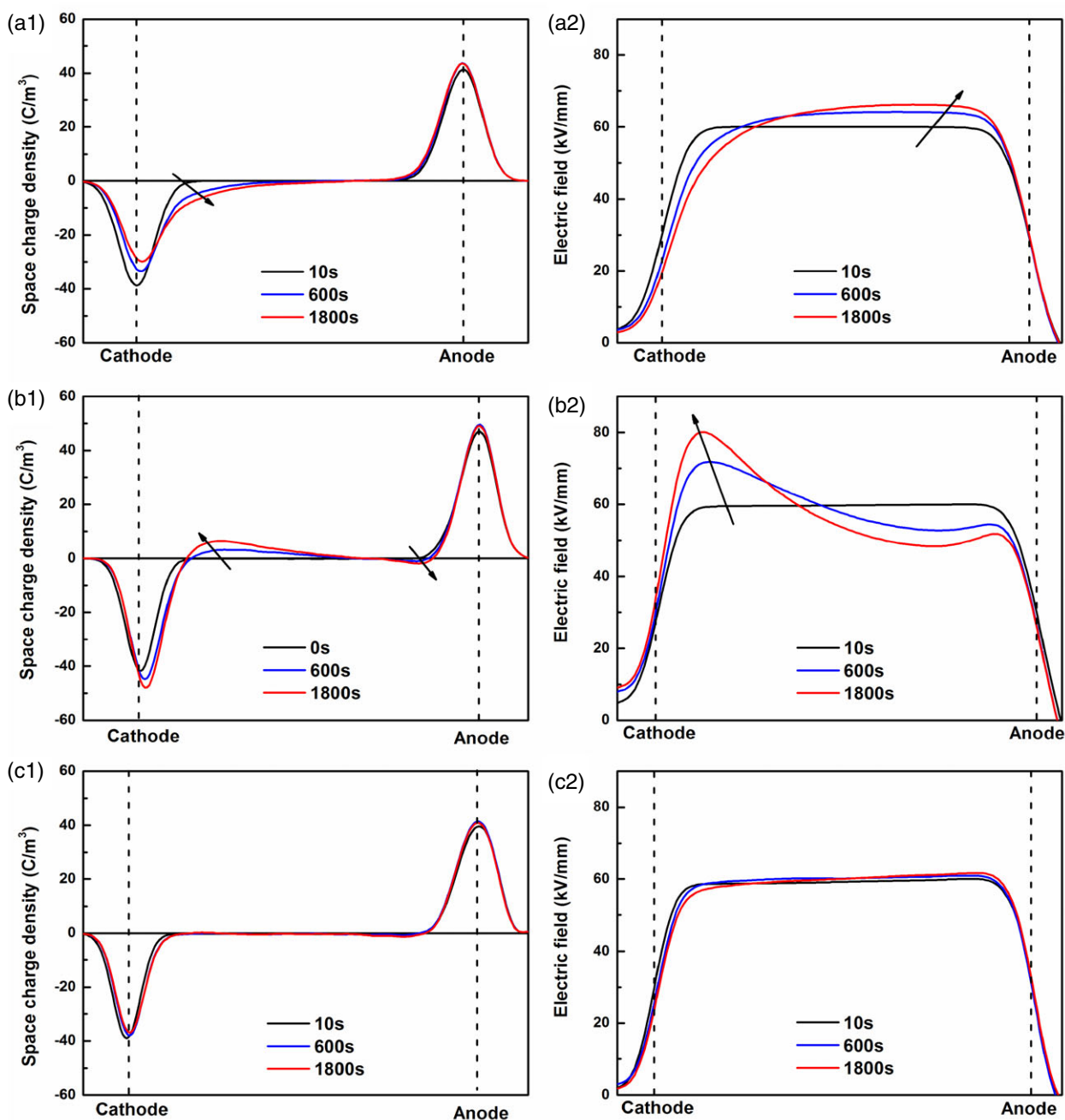


Figure 10. Space charge (1) and electrical field (2) distribution during polarization in (a) PP, (b) PP/PEC, (c) PP/EOC subjected to a dc electric field of 60 kV mm^{-1} for 1800 s.

Table 4. Summary of PEA results of PP and PP blends

	PP	PP/PEC	PP/EOC
Space charges q (1800 s) (10^{-7} C)	1.24	2.47	0.44
Maximum electrical strength E_m (1800 s) (kV mm^{-1})	66.2	80.1	61.7

suggesting that the enhancement of carrier traps in PP/EOC blends is mainly related to interfaces III. Hence, although interfaces II are responsible for carrier traps, interfaces III are the critical factor for the improvement of carrier traps.

Space charge characteristics

Figure 10 describes the space charge distribution during polarization in samples subjected to a dc electric field of 60 kV mm^{-1} for 1800 s. The distance between the two vertical dashed lines is the thickness of the sample, which is equal to the distance from cathode to anode. The total absolute amount of charge accumulated in the samples can be calculated based on the charge density distribution given in Fig. 10 according to

$$q(t) = \int_0^d |\rho(x, t)| S dx \quad (2)$$

where ρ is the charge density, t is the time, x is the spatial coordinate, S is the electrode area and d is the thickness of the sample. The electrical field distribution can be obtained from the space charge distribution by using Poisson's equation. The absolute values of space charges (q) and maximum electrical strength (E_m) in the three specimens at 60 kV mm^{-1} for 1800 s are listed in Table 4.

It can be seen from Fig. 10(a1) that a large number of homo-charges form around the cathode of the neat PP, which is the result of electron injection at the cathode. Figure 10(b1) clearly shows that a large number of hetero-charges developed near the electrodes in the PP/PEC blend. The hetero-charges in the polymer originate from the ionization of impurities mainly including additives, catalyst residues and weak polar molecules. The accumulation of positive charges near the cathode was more prominent than that of negative charges in the vicinity of the anode. This suggests that the mobility of positive carriers is much lower than negative carriers under the same electric field strength in PP/PEC blends. Figure 10(c1) shows the improved space charge distribution in PP/EOC blends. There are almost no space charges existing near the cathode and only a very small amount of hetero-space charges were observed near the anode of the samples.

The electrical field distributions are shown in Fig. 10(2). One can see from Fig. 10(a2) that the field enhancement near the cathode by homo-space charge in pure PP is ca 10.3% of the average applied field ($E_{av} = 60 \text{ kV mm}^{-1}$) after 1800 s. Figure 10(b2) shows that the electrical field between the cathode and the peak position of the positive charge in the bulk of PP/PEC blends reaches 80.1 kV mm^{-1} , which is ca 33.5% higher than E_{av} . Furthermore, as shown in Fig. 10(c2), the electrical field distribution in PP/EOC blends is ideal for HVDC cable.

Combining the results of the space charge and electrical field distributions, PP/EOC blends show excellent space charge characteristics compared with PP and PP/PEC blends. The space charge is considered to be closely related to carrier traps. PP/EOC blends

have more interfaces at the boundaries of spherulites than PP and PP/PEC blends. The shallow traps existing in the boundaries of the spherulites can act as hopping sites to improve the transportation of charges. Therefore the space charge in PP/EOC blends is remarkably suppressed.

CONCLUSION

In summary, adopting a novel elastomer, PP/PEC and PP/EOC blends were prepared. The space charges in PP/EOC blends were found to be substantially lower than those in pure PP and PP/PEC blends with the PEA tests. SEM studies show that the elastomers were distributed uniformly in the PP matrix and the PP/PEC blends have better compatibility than PP/EOC blends. TEM images clearly demonstrate the presence of an interphase layer at the PP/elastomer interface, which originates from the high compatibility between the PP and elastomers. The XRD results suggest that the crystalline form does not change after blending compared with the matrix. SAXS patterns indicate that the addition of the elastomers has no obvious effect on the lamellar thickness of the PP matrix. Thermal analysis and POM studies reveal that EOC is an effective nucleating agent, significantly increasing the spherulite boundaries, and thus the shallow traps in PP/EOC blends are dramatically increased. The improvement of the space charge distribution in PP/EOC blends is attributed to the increase of shallow traps caused by the increase of spherulite boundaries. These results also provide a simple but effective method to achieve markedly improved space charge distribution in the insulation of recyclable HVDC cable.

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